

Problem-Sized Lean Worlds

Persistent, checkable research records for AI-assisted mathematics

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ABSTRACT

A proof becomes usable by other mathematicians when it arrives with an exact statement, an explanation of its hard step, its relation to earlier work and a precise account of what remains open. This paper describes a research environment organised around a persistent unit of work, the problem-sized world: the files in a public Git repository that keep one research problem together with its results, failed routes, experiments, literature and exact open obligations. A short paper explains the strongest results to an expert, a long record keeps the complete arguments, Lean checks the formal proofs and Comparator checks selected ones against separately written statements, and a claim registry fixes the wording, evidence class and limits of every public claim. The public record supports continuation by human researchers and compatible agents beyond the originating chat or provider, and accepted work returns to it with credit. A public Lean repository maintains eight Erdős problems this way. Its results include an irrationality criterion for Problem 257, checked in Lean and by Comparator, that admits supports with divergent reciprocal sum, and a formal refutation of the Formal Conjectures statement of Problem 1041, built on a counterexample posted by the forum contributor ani; Google DeepMind's Formal Conjectures now records the answer as False and links the proof. Worked cases from Problems 257 and 249 and from a cross-problem synthesis show what the record keeps for a result, and a transfer between Problems 251 and 269, checked in Lean, shows one problem's theorem proving another's criterion and then improving its constant from $1/31$ to $1/3$. Replaying the recorded failed routes of all eight problems turned a stalled argument into a proof that the Problem 269 distinct-height sum is irrational for every set of at least two primes, which Erdős asserted without proof in 1973; Lean checks the case of the primes 2, 3 and 5. The paper compares the design with Prove2Me and blueprint-based systems and specifies the controlled comparison that will measure whether such records reduce the work of understanding, checking and continuing a result.

Contribution and scope

Design. The problem-sized world as the persistent unit of AI-assisted research; four separately recorded decisions about every returned result, on its evidence, its mathematical appraisal, its exposition and the next unit of work; a path from proof to public claim that separates what Lean settles, what Comparator settles and what a mathematician judges; and a contribution cycle that a person or an agent can run from a public clone. **Instance.** Eight Erdős problems in one Lean repository: sixteen problem papers, 1,842 Lean modules, and 695 results asserted in the papers, short and long versions counted separately, of which 674 carry a Lean declaration and 633 have been compared by Comparator. **Scope.** For Problem 1041 the Formal Conjectures statement is refuted; independent human review of its correspondence with the 1958 wording has not been recorded. The other seven targets remain open. The system leaves novelty and significance to experts.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

1 A theorem and the explanation it needs

Problem 257 asks whether $\sum_{a \in A} (2^a - 1)^{-1}$ is irrational for every infinite set A of positive integers [1]. For an integer $b \geq 2$ write $X_A(b) = \sum_{a \in A} (b^a - 1)^{-1}$, fix a finite nonempty set P of primes, and let $h(a) = \prod_{p \in P} p^{v_p(a)}$ be the P -part of a , where $v_p(a)$ is the exponent of p in a . The Problem 257 short paper [2] proves that

$$\sum_{a \in A} \frac{h(a)}{a(2^{h(a)} - 1)} < \infty$$

makes $X_B(b)$ irrational for every integer $b \geq 2$ and every infinite $B \subseteq A$. Lean proves the criterion, and Comparator has checked that proof against the separately written statement (Section 4).

The criterion reaches past reciprocal summability. Let

$$A_* = \{2^k m : k \geq 1, m \text{ odd}, m \leq 2^{2^k}\}.$$

The k th layer of A_* has reciprocal mass at least $1/4$, so $\sum_{a \in A_*} 1/a$ diverges, while for $P = \{2\}$ an element $2^k m$ contributes $1/[m(2^{2^k} - 1)]$ to the weighted sum and the whole layer at most 2^{1-k} . Every infinite subset of A_* therefore has an irrational value at every integer base. These estimates are ordinary arguments in the short paper; Lean supplies the criterion they feed.

The proof turns rationality into a lattice. For $N \geq 1$,

$$0 < \Delta_A(N) = \sum_{a \in A} \frac{b^{N \bmod a} - 1}{b^a - 1} = (b^N - 1)X_A(b) - J(N), \quad J(N) \in \mathbb{Z},$$

so if $X_A(b) = p/q$ every such displacement is at least $1/q$, and it suffices to find arbitrarily small ones. Sampling N along multiples of a modulus that every element of a finite part of A divides removes that part. The infinite tail is the hard step. Each remaining term is periodic in N ; averaging over the first T multiples of the modulus bounds its complete periods by its weighted summand when the P -part of the exponent divides the modulus, and through a large common divisor otherwise. The leftover incomplete periods of infinitely many exponents need a second average, over dyadic values of T . Its length is set between two scales. Once the finite part of A and its modulus are fixed, the sampling modulus grows polynomially in a parameter H , while the large-divisor bound has a denominator exponential in H ; an averaging length between the two makes both errors vanish as H grows (`WeightedReturn.lean` carries both averages).

Erdős proved the pairwise-coprime case of the reciprocal-summable criterion and stated, without details, that coprimality can be removed [3, p. 222]. The short paper gives a complete proof of that stated extension at every integer base, which Lean checks and Comparator has compared, and credits it to him. Erdős also remarked that reciprocal summability could be replaced by a weaker, more complicated condition, which he did not state; the short paper does not identify its weighted condition with that remark, and it credits Duverney and Tachiya for a related selection step [4]. The criterion excludes full support and the set of all odd exponents, and irrationality for every infinite support remains open.

A reader given this much can check the theorem, see why it holds, see whose idea it extends and see what is left to prove. Mathematicians writing in September 2026 describe what a proof produced at machine speed arrives without. Kra writes that the machines are producing solutions “faster than the mathematical community can read them, much less digest them” [5]; Cohn calls the unexplained remainder technical debt [6]; Tao asks for analysis that “identifies insights from the solution process” [7]; a declaration first signed by twenty-five Fields Medallists names the writeup, the isolation of new methods and ideas, and the citation of earlier work as what rushed announcements lose [8]; and Sanderson writes that “every AI-generated proof is born

an unsolved exposition problem” [9]. Each of these concerns what a proof does not carry by itself: the routes that failed, the computations that chose the route, the literature it relied on and the precise statement that remains open.

This paper describes a research environment built to keep that record. Its unit of work is the *problem-sized world*: one research problem together with everything learned about it, kept in a public repository. As producing results becomes easier, helping other people understand and use them becomes the scarce part of the work, and the world is organised around that part.

2 A problem-sized world

A problem-sized world is a versioned set of files in a public Git repository, organised around one research question and kept while the question is refined, attacked, partly answered or resolved. Its short paper, 12 to 28 pages long, is the first read: the problem and its status, the strongest results and their mechanism, their relation to prior work and the exact point where the argument stops. Its long record keeps the complete arguments, failed routes, experiments with their finite domains, corrected claims and attribution for every borrowed idea. Beside them sit the Lean source, checked by a pinned kernel [10]; a claim registry with the wording, evidence class and limits of each public claim; and a ledger recording, for every result asserted in a paper, the Lean declarations that state it or that none does. A margin mark beside a result links its declaration and any Comparator receipt; an unmarked result rests on the paper’s own argument, and a dagger marks a named input the formal proof assumes.

Every result that returns to a world raises four decisions, and the world records each one separately (Table 1). A successful check in one row settles nothing in another: a valid theorem can be a routine lemma that belongs only in the long record, a promising conjecture can direct the next computation without entering the proved claims, and a clear explanation can describe the wrong formal statement.

Problem 249 asks whether $S = \sum_{n \geq 1} \varphi(n)/2^n$ is irrational, where φ is Euler’s totient function [11]. Its short paper [12] starts from an exact theorem about the subsequences obtained by splitting the coefficients of S into residue classes modulo powers of a base. With $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\varphi(0) = 0$, the seven sections $\varphi(2^j n + r)$ with $0 \leq j \leq 2$ and $0 \leq r < 2^j$ reduce to the five sequences

$$\varphi(n), \quad \varphi(2n), \quad \varphi(2n+1), \quad \varphi(4n+1), \quad \varphi(4n+3),$$

since $\varphi(4n) = 2\varphi(2n)$ and $\varphi(4n+2) = \varphi(2n+1)$. These five are linearly independent, and the pattern holds in every base. For every base $k \geq 2$ and level $e \geq 1$, the sections $\varphi(k^j n + r)$ with $0 \leq j \leq e$ and $0 \leq r < k^j$ span a space of dimension $k^e + 1$ over $\mathbb{Q}$, with basis $\varphi(n)$, $\varphi(kn)$ and the sections with $1 \leq j \leq e$ and $k \nmid r$; every other section is an explicit integer multiple of a basis element, and these reductions generate all integral relations. Coons proved that φ is not k -regular for any $k \geq 2$ [13]; the basis theorem recovers this and gives the dimension at every level.

Lean proves the theorem for every base, and Comparator has compared it. The reductions are elementary; the independence carries the weight, through an evaluation matrix made diagonal and nonzero modulo an auxiliary prime, with evaluation points chosen by the Chinese remainder theorem and Dirichlet’s theorem so that each row asks one primitive affine form to take a prime value. Martin’s Theorem 1 already implies the affine independence [14], and the separation method goes back to Yazdani, who credits it to Shallit [15]; the short paper credits both. The record keeps the basis theorem, a statement about the coefficients, apart from the irrationality of S , whose open form it states exactly.

Write $R_N = \sum_{j \geq 1} \varphi(N + j)2^{-j}$ for the tail of S after N terms, put $H_t = \text{lcm}(1, \dots, t)$, and write $\text{Cert}(t)$ for a finite exact computation, on totient values just after H_t and $2H_t$, showing that

Decision	What the world records	Weighted criterion	Dyadic tail differences
Evidence	The exact statement, its hypotheses and named inputs, the Lean declaration and its checking receipts	Lean proves the criterion; Comparator has compared the declarations that state it	Lean proves the equivalence with $1/31$ and then with $1/3$; the application trace records the rejected and the accepted step
Appraisal	The mechanism, the closest antecedent, the strength of the result and the obstruction that survives	Two averages; Erdős's reciprocal-summable case, credited; admits A_* ; excludes full support	A change of state variable; $1/31$ was inherited from radix 30, and $1/3$ cannot be raised for radix-two orbits in general
Exposition	Whether the result belongs in the short paper or the long record, and how its hard step is explained	The theorem opens the short paper; the example and the full averaging proof follow it	Section 3 shows the whole transfer; the experiment keeps the trace
Allocation	The next question, the test that would decide it and the condition for stopping	Irrationality for every infinite support stays the registered endpoint; a changed-base reader exercise and independent replay route now test outside use of the weighted condition	Is $1/3$ best for irrational tails? A probe pointed to 0.4125, the literature fixed the answer, and the question closed. The restricted-shift theorem and reader exercise now test whether an outsider can use that answer; outside use has not yet been observed

Table 1: Four decisions about a returned result, for the weighted criterion of Section 1 and the cross-problem transfer of Section 3. An enduring endpoint and a justified next step are different allocations.

$R_{2H_t} - R_{H_t}$ is not an integer. Lean checks, and Comparator has compared,

$$\forall t \leq 82, \quad \text{Cert}(t),$$

and Lean also proves, again with a Comparator comparison, that irrationality of S is equivalent to the unbounded supply

$$\forall T, \exists t > T, \quad \text{Cert}(t).$$

The finite theorem makes no $t = 83$ or cofinal claim. What remains open is the unbounded supply itself, which is the irrationality of S in equivalent form, and the record states it in exactly that form.

3 From corpus to insight

The short papers are distilled from a much larger research record. Each problem keeps a research corpus of exact results, negative results, computations, literature notes and proposed next steps, including intermediate and superseded entries, and the mathematical loop runs over it in two coupled roles.

Discovery and stewardship. A discovery pass stays close to the frontier: it runs discriminating computations, attempts proofs and returns results. A stewardship pass reads each new result against the whole corpus and makes the four decisions of Table 1: what the result establishes, whether it is routine or strong and whom it credits, where it belongs in a paper, and which obligation deserves the next unit of work. One agent may play both roles in sequence. Neither inherits the other’s authority. Stewardship runs after a stable result or review event, and discovery resumes when the recorded frontier justifies a further discriminating attempt; an unchanged record is no reason to poll or repeat a run.

Models do the reading, comparison and drafting under written procedures, published with the repository as agent skills. Deterministic code assembles each long record from its authored parts, records the Lean evidence of every asserted result or its absence, and checks that each result a paper states is linked and that each registered limitation stays visible. No deterministic check decides what is important; that judgement, whoever drafts it, is written into the paper, where a reader can inspect and dispute it.

Testing a proposed measure of value. Patel et al. [36] propose conditional proof difficulty as a way to choose useful mathematics: a lemma is valuable when making it available shortens later proofs. Their intrinsic score divides proof cost by the length of the statement and its required definitions. Their direct-use utility multiplies that cost by the number of citing theorems. This gives stewardship a testable proposal, but the shared proof-cost factor means that correlation between the two scores alone cannot establish that either finds useful abstractions. Our [accompanying experiment](#) compares the observed correlation with shuffled user counts, separates generated certificates and repeated statement text, and records the resolved part of the dependency graph. It uses indexed subtotals of recorded source proofs; it neither trains their value model nor estimates a shortest proof.

On the source snapshot used here, the two proxies have Spearman correlation 0.799 over 9,289 cited theorem or lemma declarations. Yet their common proof-cost factor survives shuffling the user counts: all 64 fixed-seed permutations give a larger correlation, with mean 0.833, while the interestingness proxy correlates -0.172 with user count alone. This is a negative control against interpreting the local correlation as independent evidence of reuse prediction, not a test of the authors’ trained model. Excluding every target with generated-certificate ancestry leaves 8,022 declarations and correlation 0.856; excluding targets whose indexed dependency closure reports any omitted internal reference leaves 2,417 and 0.660. These are different selected populations, not causal estimates. The experiment therefore reports source coverage alongside each result: the index resolves 16,111 declarations and omits private helpers and some public dependencies. Its costs are subtotals in that index. The analysis turns an apparently reassuring correlation into a concrete demand for independent evaluation.

Conditioning on a workload gives a different question. In an [eight-target inspection experiment](#) drawn from the Problem 251 and 269 papers, shared prerequisites account for 3,529 indexed lines, compared with 6,135 if every target is inspected separately. Among the same candidate premises, the leader by use count times cost removes only 30 lines from the shared subtotal; the largest initial contextual cut removes 491. Neither number determines importance: reading the source identifies the latter as an equivalence wrapper and redirects attention to the underlying arguments. Two Problem 269 premises remove 220 and 216 lines separately, but 467 together, because a shared dependency stops being needed only after both paths to it are cut. Thus contextual gains can increase; adding individual scores misses this complementarity, and greedy selection has no general optimality guarantee. The paired tests make the workload and available premises explicit before allocating an inspection.

The resulting candidates carry different jobs. A widely reused lemma needs an explanation of the mechanism its users share; a repeated statement needs a comparison of its elaborated types and the pinned library; a large certificate dependency needs an inspection of the generator before

its size is read as mathematical difficulty. A negative result that eliminates an unproductive attack can deserve attention despite few formal users. The [stewardship procedure](#) requires a named consumer or falsifiable test and records the allocation separately from the score. Such a ranking changes which question is checked next; the returned evidence determines whether the appraisal changes.

Negative results are part of what the loop produces. Tao observes that withheld negative results and processes hide where a new frontier lies [7]; the record keeps them. A counterexample refutes a conjecture on its exact domain, and a Lean no-go theorem rules out a class of strategies under explicit hypotheses, so later work need not repeat either. A failed computation records its finite domain and the inference it defeats. For Problem 257, a proposed estimate said that the greedy test ruling out a rational number with denominator Q as the sum of an infinite subseries of $\sum_n 1/(2^n - 1)$ stops within about $2\log_2 Q - 3.3$ steps; an [exact rational probe](#) first rules out 189/388 at step 17. The record keeps both facts: the estimate is false, and the problem itself is untouched. Inside the weighted proof the same discipline explains the second average: the short paper shows why the majorant of the reciprocal-summable proof fails beyond that class.

Replaying the record. Zheng et al. [25] store a completed discovery run as a tree of attempts with their evaluated outcomes, so that another exploration policy can be replayed over the tree without rerunning any attempt; a separate agent improves that policy while the discovery agent and evaluator stay fixed, and a route outside the recorded branches earns nothing. In a problem-sized world the reader of the record is the agent that makes the next attempt, and the [proof workbench](#) records moves and kernel verdicts but by design does not search, rank or decide. The record also keeps two kinds of outcome: a checked theorem or an exact computation replays identically and is a premise in every world, while a failed attempt closes nothing until its cause is located. On 26 September four readers replayed the recorded routes of the eight worlds under that distinction ([replay](#)), classing 70 decisive rows as proofs, obstructions, counterexamples, finite computations, stalled attempts, superseded formulations or routes never tried; a checker rereads every cited record at the pinned commit. Three reopened routes became theorems. The Problem 251 record had tried to force a repeated tail under rationality and stalled because its tails are unbounded. In the distinct-height sum of Problem 269, where each running least common multiple counts once, grouping the terms into dyadic shells makes the tails bounded and uniquely decodable, and the same return argument proves $\mathcal{D}_{\{2,3,5\}}$ irrational [31], and Lean checks the proof ([DistinctHeightIrrationality.lean](#)). Erdős asserted irrationality of these sums in 1973 without printing an argument [32]. The general case came from restating the sum in another field's terms, as a Cantor series whose bases follow the cutting sequence of the flow $t(1, \dots, 1)$ on $\prod_{q \in P} \mathbb{R}/(\log q)\mathbb{Z}$; Kurgan et al. find that the lemmas a proof uses rarely share its statement's vocabulary [30]. Rationality forces two tails taken at a close return of the flow to be equal, and an Euler-characteristic count on a subtorus supplies a return at which they differ by one transposition, so $\mathcal{D}_P$ is irrational for every set P of at least two primes. For two primes this already follows from transcendence of Hecke–Mahler values [26, 27]; a search in both vocabularies found no result for three or more. The attack on the full catalogue sum failed, and the failure left a theorem: with the carry f replaced by the product of the primes that follow p in a chain, the same comparison proves irrational every sub-sum E_p over the powers of one prime p , and for two primes the catalogue sum factors exactly as $E_p E_q$. Axelrod et al. mine the lemmas of failed proof searches for abstractions and keep one only when it speeds a later proof [29]; here the kept object is an operation defined by what it preserves, and its consumer is a theorem. The synthesis paper had proved transcendence along eventually doubling chains with Mahler's method and recorded chains at base $4/3$ whose ratios are 2 except for infinitely many 3s as its first open case; Schlickewei's Subspace Theorem in S -unit coordinates [33, 34] proves transcendence for every infinite divisibility chain at every rational base $a/b > 1$ [16]; integer bases already follow from Roth's theorem and Mahler's method, so the new cases are non-integer bases with bounded

aperiodic ratio words. Erdős’s 1948 divisibility argument, which the Problem 249 record had stopped on coefficient size, proves irrationality once the totient coefficients are reduced modulo slowly growing powers of two, a statement that a 2026 criterion of Kaneko, Suzuki and Tachiya also implies [28]. The general case, the sub-sum theorem and these last two proofs are ordinary arguments, the first two checked step by step by a second, independent agent, awaiting specialist review, and the replay compares no exploration policies.

Reading the problems together raises questions that none of them asks alone. The smallest complete case is a transfer between two problems. For dyadic tails $T_{N+1} = 2T_N - g_{N+1}$ with integer digits g_N , the Problem 251 record proves that T_0 is irrational exactly when no difference $T_{N+h} - T_N$ with $h > 0$ is an integer. The Problem 269 record proves an escape theorem: an orbit $x_{n+1} = p_n x_n - c_n$ with $p_n, c_n \in \mathbb{Z}$ and $2 \leq p_n \leq 30$ reaches an integer or returns infinitely often to distance at least $1/31$ from the integers. Applied to T , the escape theorem says nothing about differences. The difference $U_N = T_{N+h} - T_N$ satisfies $U_{N+1} = 2U_N - (g_{N+h+1} - g_{N+1})$, so it applies to U , and the Problem 251 criterion excludes the integral alternative. Hence T_0 is irrational exactly when every such difference is at least $1/31$ from the integers for infinitely many N . The `experiment` imports only the two sources and records the Lean outcomes: the application to T is rejected, the application to U leaves one goal, and the Problem 251 theorem closes it. The change of variable is the mathematical bridge, and Lean establishes what follows. The constant $1/31$ came from the radix bound 30. With the bound B as a parameter the same proof gives $1/(B+1)$, which the orbit alternating between $1/(B+1)$ and $B/(B+1)$ shows cannot be improved in general, and hence $1/3$ for the differences. Lean checks all three, and the $1/31$ run stays in the record as the original outcome. A finite probe then asked whether $1/3$ is best for irrational tails, and a published theorem answered it. Dubickas proved $\limsup_n \|2^n \xi\| \geq \tau$ for every irrational ξ , with equality at the Thue–Morse number $\tau = 0.412454\dots$, as stated by Akiyama and Kaneko [46], so every fixed $0 < c < \tau$ works in the equivalence. The endpoint does not: the doubling orbit of τ stays strictly closer than τ to the integers at every positive index, so the admissible constants form the open interval $(0, \tau)$. The record keeps the Lean-checked $1/3$, the cited theorem and the ordinary endpoint argument, in the `recorded closure`, at their own evidence classes. This sharpens a quantitative irrationality criterion; establishing the criterion for the actual prime gaps remains open.

The transfer also makes conditional proof cost concrete. Holding the target fixed, one can inspect the source work counted when the recorded proof is expanded through the shift criterion or the escape lemma, and the work counted when these are treated as available premises. Patel et al.’s composition inequality bounds ideal proof cost through an intermediate lemma; their Bellman-type equality requires an additional optimal-intermediate assumption [36, Definition 1]. A dependency expansion of these recorded proofs is an inspectable accounting model, not evidence for that assumption. The successful change from T to its difference U remains the mathematical reason these two premises combine.

The sharp-constant question then leads to a question about the shifts being tested. A rational dyadic orbit is eventually periodic modulo one, so a family containing a multiple of every positive integer detects its eventual integral differences. The `synthesis theorem` proves the converse with explicit rational counterexamples: factorial shifts suffice, while power-of-two shifts do not. The `reader task` asks which shifts still work and why a limsup bound does not include its endpoint. The full restricted-shift classification is an ordinary argument using the cited Dubickas bound; the Lean evidence currently covers the $1/3$ all-shifts criterion.

Seven of the eight problems ask whether a series is irrational, and the synthesis paper [16] studies what their methods share and where each stops. The sparse corrections used for Problem 251 led to a capacity criterion for factorial digits, and then to a question: can one sequence of factorial digits prescribe several derivatives at once? For $c > 0$, let $\mathcal{H}_c$ be the class of entire functions $f(z) = \sum_{n \geq 1} e_n z^n / n!$ with $e_n \in \mathbb{Z}_{\geq 0}$, $e_n \leq n^c$ eventually, and every fixed positive

integer dividing e_n for all large n , and for an integer $d \geq 1$ put $J_d(f) = (f(1), f'(1), \dots, f^{(d-1)}(1))$. The synthesis paper proves

$$\dim_{\mathbb{H}}\{J_d(f) : f \in \mathcal{H}_c\} = \min(c, d),$$

that this set contains a nonempty open subset of $\mathbb{R}^d$ exactly when $c > d$, and that a nonpolynomial member of $\mathcal{H}_c$ with $f(1), \dots, f^{(d-1)}(1)$ all rational exists exactly when $c > d$. So a nonpolynomial $f \in \mathcal{H}_2$ with $f(1)$ rational has $f'(1)$ irrational, while coefficients of size $n^{2+\varepsilon}$ allow both to be rational. The construction rests on a carry identity,

$$(z-1) \sum_{n \geq 0} \frac{b_n z^n}{n!} = -b_0 + \sum_{n \geq 1} \frac{(nb_{n-1} - b_n) z^n}{n!} :$$

adding such a term to f leaves $f(1)$ unchanged and moves $f'(1)$ by $\sum_n b_n/n!$, and a factor $(z-1)^k$ leaves $f(1), \dots, f^{(k-1)}(1)$ unchanged at a cost of k powers of n in the coefficient allowance. Division by $(z-1)^d$ gives the matching obstruction: vanishing derivatives become bounded integer coefficients, which eventual divisibility forces to vanish. Both theorems are proved in the synthesis paper by ordinary argument and carry no Lean mark; Lean checks the finite carry identity and its preservation of divisibility.

The same corpus reading can suggest a representation absent from the recorded dependency graph. For a divisibility chain of exponents, the synthesis paper previously used a Mahler equation only when successive ratios eventually equal two. Grouping a repeated block of integer ratios instead proves that *every eventually periodic ratio chain has a transcendental Lambert subsum at every algebraic real base greater than one*. For alternating ratios 2, 3, the support is $\{6^k, 2 \cdot 6^k : k \geq 0\}$ and its generating function satisfies

$$G(z) - G(z^6) = \frac{z}{1-z} + \frac{z^2}{1-z^2}.$$

The old doubling equation fails at coefficient four. The block decomposition proves the new equation at every degree; infinitely many boundary singularities and Nishioka's value theorem supply transcendence. This covers base 4/3, where the earlier denominator estimate fails. The full ordinary proof and attribution are in the updated synthesis theorem [16]; the [coefficient probe](#) also rejects the same block equation on a specified nonperiodic ratio word. Finite agreement is a test of the proposed representation, not the infinite proof, and arbitrary nonperiodic chains remain beyond this argument. This is an application of classical Mahler theory, with no Lean or novelty claim. It illustrates a candidate-generating move—search for a functional equation before improving the estimate—without attributing the result to a numeric ranking or claiming measured discovery advantage.

4 From a proof to a public claim

A proof becomes a public claim by crossing three questions that the word “verified” tends to merge: whether the proof establishes its formal statement, which the pinned Lean kernel answers in continuous integration; whether the proved statement is exactly the separately written one, using only permitted axioms, which Comparator answers for selected statements; and whether the formal statement says what the problem says and whether the result matters, which a mathematician answers.

Lean verifies that a proof establishes the formal statement written in the source; it does not verify whether that statement captures the intended mathematics or whether the paper describes it well. The gap is measurable. Feng et al. had mathematicians grade 200 candidate solutions to Erdős problems produced by Aletheia, an agent built on Gemini Deep Think: 63 were technically

correct, of which 13 were meaningfully correct; the other 50 answered a reading of the problem that missed Erdős’s intent [17].

Comparator, a checker the Lean FRO built for judging machine-written proofs [18], answers the second question. A trusted challenge module restates each statement without its proof; the proof must be a term of exactly that type, using only the permitted axioms, and the Lean kernel must accept it. The corpus replay, run in the companion repository *plectis-erdos-lean*, also passes every compared proof through *nanoda*, an independent kernel [19]. In this repository, continuous integration also requires Comparator to reject a deliberately altered statement. This control exercises that mismatch path; the surrounding invocation and verdict contract have separate obligations. Comparator cannot decide whether a statement is the right translation of the problem. It guarantees that the separately written statement, which a reader can inspect, is the statement that was proved, so “Comparator-checked” is accurate where “independently verified” would overclaim.

The paper-to-Lean ledger makes both formal questions countable for the papers’ own statements. It holds one row for each result asserted in the sixteen problem papers, 695 in all, counting a result stated in both a short paper and a long record twice, with its statement digest and kind of correspondence. For 508 rows the named Lean declarations state the result exactly; for 141 they state it or a result that implies it by an immediate specialisation; 25 are proved from a named input, marked with a dagger; and 21 have no Lean statement. Comparator has compared 633 of the 649 exact and specialising rows, and the other 16 are queued for its next replay; every receipt in the pinned corpus replay was accepted by both the Lean kernel and *nanoda*. A digest detects a changed statement, and the classification of a correspondence stays the maintainer’s judgement.

The third question stays with people. The claim registry records the wording of each public claim with its evidence class, the bounded domain of each finite instance and the open statements it approaches, and the papers, README and generated views may say only what the registry allows. In this repository the maintainer is responsible for that wording, and no independent mathematical review is yet recorded. A release checker keeps those surfaces consistent with the registry and the Lean source, and requires a registered limitation to stay visible wherever its claim appears.

That requirement comes from a test in which a limitation escaped. One deliberately false README edit changed a clause saying that the finite cases of Problem 249 did not supply the open requirement into one saying that they completed it. Lean was untouched, and the checker passed because that relationship had not been registered. After registration, the post-repair witness accepts the current README and rejects a test copy containing the false clause. A small historical study tested the checker with deliberately false edits: nine of the ten edits were rejected. One escaped, for the reason just given. The edits were authored by the checker’s author. The original run logs were not retained. The other nine edits were not rerun against the extended checklist. The evidence marks a coverage boundary, not a reliability score.

The workflow does not technically force a second independent mathematician: the maintainer can edit a Lean statement, its record and its prose together so that every comparison agrees with the same mistake. Independent review is recorded as an external event when it happens. A changed result reopens what depends on it. Propagation starts from the changed object and lists every reverse import, claim record, paper, experiment and open obligation the change may affect; each must be updated, verified unchanged, deferred with a reason or marked out of scope, and an empty search never counts as evidence of no consequence.

5 The contribution cycle

Someone outside the original research can contribute a correction or take an argument further, and that work returns to the same record. The public repository publishes this research, checking and revision cycle as thirteen agent skills that install into common agent harnesses. Each skill owns one job, with its starting state, the changes it may make, the evidence it must return, its stopping condition and the next owner. The ordinary path orients (explain-public-system), works on one open obligation (mine-open-problem), validates (land-lean-proofs), propagates (propagate-research-consequences), packages the return with its prose held to the evidence (public-mathematical-writing) and proposes a pull request from a named starting commit (submit-pull-request). The [companion contribution protocol](#) gives the full sequence, and the [companion runtime paper](#) describes navigation and proof receipts in a fresh clone. Adoption replays the contributor’s original change from its named starting commit before reconciling it with the current tree, so the original work and any conflict resolution are credited separately. A cycle closes when the change has evidence and every consequence has a recorded disposition; a problem closes only when a proof or a refutation settles its original statement.

Credit. The [credit policy](#) lets an idea be credited separately from its proof, its Lean formalisation and its explanation, the roles Kra asks journals to distinguish [5], and credits software as a role of its own. A first return can be a plain-language research-progress issue or email: no clone, proof, code or receipt schema is required. Accepted work receives a public receipt tied to exact artifacts. Acceptance does not establish theorem status, novelty, or release inclusion. Corrections append to the history, and an earlier contributor’s record stays in it. Work also leaves the repository: Google DeepMind’s Formal Conjectures maintainers have merged four of its contributions, linking formal proofs for Problems 257, 1049 and 1041 and a constant-base variant of Problem 258.

The public [contribution record](#) contains one accepted architecture return from 23 September: OpenAI Codex repaired the ordinary-language route into the research-return workflow and supplied a checked-positive architecture packaging case. The focused checks were reproduced by the same contributor in a maintainer-operated review. This is evidence that the credited local adoption protocol has been used, not an independent outside clone replay or mathematical endorsement.

5.1 Worked example: Problem 1041

Problem 1041 shows the cycle end to end, with the decisive idea arriving from outside. It asks whether, for a monic polynomial f with all roots in the open unit disc, two roots can always be joined inside the lemniscate $\{|f| < 1\}$ by a path of length less than 2 [20, 21]. On 7 September 2026 the erdosproblems.com contributor ani posted a degree-seven counterexample, found, as the post says, with the help of GPT-6 [22], and the record took the construction as an attributed input five days later. Lean first checked the total-variation form. On 22 September it proved that for one polynomial of that construction every preconnected subset of the strict lemniscate containing two distinct roots has one-dimensional Hausdorff measure greater than 2 ([HausdorffLength.lean](#)), which refutes the exact path-image-length statement in Google DeepMind’s Formal Conjectures repository [23]. On 23 September the Formal Conjectures maintainers merged a change that marks the problem solved with answer False and links this proof [24]. The short paper and long record carry the proof with its Lean mark and credit ani for the construction. One judgement remains open: whether the formal statement matches the 1958 wording [21, Problem 5, p. 139]. The record says so beside the result.

6 Eight problems in one repository

The [public repository](#) maintains eight Erdős problems [1] as problem-sized worlds. Table 2 gives one result for each problem and the point where it stops.

Problem	A result to start with	Where it stops
#68, factorial denominator	Lean checks that the logarithm of the uncleared common denominator grows at least like $N^{3/2} \log N$, and that irrationality is equivalent to cofinally many non-unit factorial carries	Those carries are not produced
#243, Sylvester-tail rigidity	For a strictly increasing sequence of positive integers, Lean checks irrationality of $\sum 1/a_n$ under the exact cubic rate $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$	The unrestricted Sylvester-tail question
#249, binary totient series	For every base $k \geq 2$ and level $e \geq 1$, the totient sections $\varphi(k^j n + r)$ with $0 \leq j \leq e$ and $0 \leq r < k^j$ span a space of dimension $k^e + 1$ over $\mathbb{Q}$; certificates for every $t \leq 82$	Certificates for arbitrarily large t
#251, prime-gap dyadic series	Lean and Comparator check that sparse corrections preserving the congruences fill an interval of sums; from the prime number theorem and a lemma of Schläge-Puchta as named inputs, Lean checks a sequence with the local statistics of prime gaps and a rational dyadic sum	An argument specific to actual prime gaps
#257, infinite exponent supports	The weighted-support criterion of Section 1, at every integer base	Arbitrary infinite supports
#269, three-prime running LCMs	Lean and Comparator check that the three-prime running-LCM kernel has nonsingular minors of every order, so no finite sum of separated products represents it; both two-prime sums are transcendental by a cited Hecke–Mahler theorem, the repeated-sum case first posted by Fan [35]	The three-prime case
#1041, lemniscate connections	The Formal Conjectures statement is refuted with ani’s example	Match with the 1958 wording
#1049, rational-base Lambert series	Lean checks irrationality in Zudilin’s rational-base region and exact Hankel orders	Base 3/2 and all rational bases

Table 2: One result per problem, with its stopping point. The papers state each result in full with its hypotheses and sources.

The repository holds 1,823 Lean modules, counting the two library root files, under a pinned toolchain. The root-import check counts the 1,821 modules beneath those roots. It also holds sixteen problem papers, a cross-problem synthesis paper and three systems papers. The claim registry holds 150 public claims in seven statuses, from proved here and formalised here to conditional reduction and open, and states 19 remaining obligations exactly. Continuous integration builds the supported Lean roots, a coverage build compiles every module a paper cites, and

the release checker checks claims, papers, generated views and licences. A fresh clone needs no private file, and no public theorem depends on an unpublished lemma.

The clone also carries the machinery for continuing a proof. A `proof-state compiler` takes the declarations that static queries nominate for a goal and asks the pinned Lean environment what each application does. A transition exists only when Lean reaches the state after the application; the subgoals Lean leaves form an AND set that becomes the exact remaining obligation, competing applications are OR alternatives, and a proof receipt exists only when Lean exits with no goals left. In the transfer of Section 3 both applications end with no remaining goals, the rejected one because Lean reached no state after it, so acceptance and closure are recorded separately.

The records come from a private environment, built over the past year, that runs agents from several model providers, including Anthropic’s Claude and OpenAI’s GPT models, concurrently on one shared file system. Durable state lives in files, and before an agent opens a large source a router compiles a packet conditioned on its task, with each item labelled by evidence class and a list of what was omitted. Agents claim exact paths for a bounded lease before changing them. Lean builds pass through a semantic single-flight queue keyed by source, targets, toolchain and dependency lock, so equivalent requests share one build, and a deferral is recorded separately from theorem failures. A failure in navigation, scheduling, experimentation or validation can change a route, skill or check for later agents through a guarded proposal, and it never changes the status of a mathematical claim. The public repository replays without the private environment: the private environment explains how the records were produced, and the public evidence is what supports each claim.

7 Relation to Prove2Me and other systems

Prove2Me [37] is the closest published system: a hosted platform on which anyone with a coding agent states Lean theorems and submits proofs. Each statement is immutable and may carry several proofs; the server accepts a proof only when its type is exactly the target’s, with no sorry and only whitelisted axioms, and a disproof proves the negation. Missions on papers, textbooks and open problems confine human audit to a goal, its definitions and milestone lemmas, aided by a blind agent read-back of each Lean statement, and a proof sketch may import open statements for other agents to close. Its published examples also include proof explanations, discussion and a formalisation corrected after a disproof; its service documents statement deprecation and relinking. Persistence, correction, shared work and exact statement checking are therefore overlapping capabilities. The comparison is about what a reader can do with each public research object (Table 3). Prove2Me’s paper appeared on 28 August 2026; the first public version of this paper, with its claim registry and release checks, appeared in July 2026, and the two designs developed independently. Prove2Me is the published reference for hosted statement and proof separation, audited mission cores and read-back auditing.

The two can compose. Prove2Me provides hosting, importable statements, per-submission checking and a contributor base. This repository offers a particular problem-level record around a statement: papers, failed routes, obstructions and open obligations. Whether that record makes a changed hypothesis easier to understand or a returned result easier to review is an empirical question, not a feature-exclusivity claim. This repository’s sharp constant in the collinear root-segment bound of Erdős, Herzog and Piranian for Problem 1041, which Comparator has also compared, is hosted on Prove2Me as a package of 21 theorems with accepted proofs, ported from Lean 4.29.1 to Prove2Me’s Lean 4.30.0 environment.

Other systems. Feng et al. ran an agent over the 700 Erdős problems then listed as open, published graded outcomes [17], and propose reporting AI-assisted results on at least two axes, autonomy and significance, with significance judged only by mathematicians expert in the

	Prove2Me	Problem-sized worlds
Unit of work	A mission on a paper, textbook or open problem, built from immutable statements and proofs	One research problem with its whole record
What persists	Statements, proofs, sketches, explanations, milestones and discussion; disproofs and deprecations stay visible	Short and long papers, failed routes, experiments, typed claims and open obligations beside the Lean source
Exact-statement check	Every submission, on the server	Selected statements, through Comparator, with a second kernel in the corpus replay
Statement fidelity	A captain audits each mission core with a blind read-back; a moderator approves public missions	The maintainer is responsible for the wording of each public claim in the claim registry; no independent review is recorded yet
Exposition	A description for each statement and an explanation for each proof	A short paper and a long record for each problem, with evidence marks at each result
Credit	Permanent names, trust scores, leaderboards and import citations	Receipts by contribution role, and attribution for outside ideas
Contribution path	Web platform and API, with an account	A plain-language issue or email without a clone; a clone and agent skills for source changes

Table 3: Prove2Me and problem-sized worlds, from Prove2Me’s paper and public pages and from this repository, on 24 September 2026.

area [38]; the claim registry leaves significance to experts in the same way. Tao’s community wiki listed AI contributions to the Erdős problems, failed attempts included, until its updates stopped on 30 June 2026 [39]. Formal Conjectures [23] turns open problems into formal targets, and Ringer distinguishes mathematics as a process from mathematics as a benchmark [40]; a world keeps the process around such a target. Blueprints connect informal proof plans to Lean declarations [41]; LeanArchitect extracts blueprint data from annotated Lean source, infers dependencies and tracks progress [42]; and LeanMarathon maintains an evolving blueprint, one Lean file serving as proof skeleton, natural-language proof graph and shared record, worked by separate construction, audit, proof and repair agents [43]. These are direct precedents for durable formal development linked to its explanation, and a world can use a blueprint as its dependency map; the world adds the appraisal around the proofs, with the mechanism and antecedents of each result, the failed routes and the exact open obligations. Hattori, Matsuzaki and Fujiwara translate Lean proofs into natural language step by step and summarise along the proof structure [44], a direct precedent for the explanation a world keeps beside a formal proof. Each component exists elsewhere; a world contributes their combination on one task, as in the transfer of Section 3, where the record selected the move, Lean checked it, the explanation sits beside the proof and the next question carries its test. The measurement of Section 8 compares that workload with the alternatives. Provers such as ReProver, built on LeanDojo, and Open-

Prover [47, 48] can supply a world with candidate proofs, and Li et al. hold generation and validation apart in research mathematics without a proof assistant [49].

8 Limits, and what stronger models change

The other seven targets remain open, and independent human review of the correspondence between the Problem 1041 refutation and the 1958 wording has not been recorded. The system records evidence classes and review events; novelty and significance are judgements it leaves to experts, as Feng et al. also propose [38]. The examples here show how the record organises results; whether the workflow made those results more likely has not been measured. By 24 September 2026 no outside human contributor had opened a pull request or issue in the repository; its outside record is the Formal Conjectures merges of Section 5 and the Prove2Me package. A record also leaves several of the concerns in Section 1 where it finds them: it trains no one, it restores no signal of individual expertise [50], and it answers neither the ethical objections some mathematicians raise to AI-assisted mathematics [51] nor the warning of forty-two Fellows and Foreign Members of the Royal Society about catastrophic risk from AI [52].

The same design serves stronger models. A stronger model starts from the current frontier with the failed routes already marked, and it can reorganise, re-explain and extend the record as well as the mathematics. The public interfaces admit any producer: an outside contributor can attach any agent runner to the clone, and a laboratory with many frontier models could replace the private production environment while keeping the same evidence boundary.

Gowers suggests a well-designed database of what is known, probably built with AI help, and adds that it may prove unnecessary if one can ask a language model for a bird’s-eye view of an area [53]; Koukoulopoulos proposes a public facility in which coordinated agents explore a research programme divided into subproblems under researchers’ direction [54]. A problem-sized world is one concrete way to maintain the sources, explanations and open questions such work draws on, here at the scale of eight problems, and the measurement below tests its value against an on-demand briefing built from the same material. Antieau writes that texts generated largely by a language model deserve a home other than the arXiv [55]; a world gives its long records one beside the proofs they explain. The same structure suits a single landmark result, and it asks three things of a field: stable result identities, a checker where one exists, and an explicit boundary on public claims.

The next measurement is comparative. With the source revision, task set, model, tool access and budget held fixed, agents receive the Lean source alone, an information-equivalent written briefing, or the whole world with its continuation tools. A representation contrast holds information, tools and budget fixed and varies only the organisation and recorded decisions; a product contrast compares the world with the best on-demand briefing and tools built from the same material, with preparation amortised over observed reuse and maintenance, review and repair counted on both sides. miniCTX’s repository contexts and temporal splits are a precedent for these controls [45]. Independent domain readers grade whether each recovers the strongest supported claim, its hypotheses, its hard step, its antecedents and the remaining obligation, and review one returned change; statement errors, reviewer time and total work, preparation and maintenance included, are recorded, and failed and abandoned attempts are counted. Ablations separate the short paper, the evidence links and the continuation tools; historical tasks need a cutoff that hides later proofs and theorem names, and prospective tasks give a separate test. Tasks separating meaning from wording include a new antecedent that leaves the Lean proof intact and changes the appraisal, as the Dubickas comparison did, a dropped hypothesis or a limsup bound read as a bound reached infinitely often, and a faithful paraphrase. The transfer of Section 3 is a diagnostic fixture; discovery comparisons use held-out transformations or

prospective continuations. Repeating the frozen tasks with a stronger model measures what the record adds as models improve.

For score-guided allocation, freeze the candidate statements and their rankings before the continuation period. Compare with proof-size, direct-use and unranked selection under equal effort, recording later independent consumers, accepted reuse, review cost and abandoned attempts. Scoring the selected outputs by the selection score would test the optimiser rather than the usefulness of its choices.

9 Conclusion

A problem-sized world keeps the record that lets other people understand a proof, check it, credit it and build on it: its mechanism, its antecedents, its formal evidence and its exact open statements. The weighted criterion, the totient basis and the derivative-interpolation theorem show that record for three results, the dyadic transfer shows the corpus supplying a proof its parts did not contain, and the eight worlds in the public repository show that it can be kept persistent, checkable and open to contribution across several hard problems. The record is built for models stronger than the ones that produced it: each starts from the recorded frontier with the failed routes marked, and Section 8 names the measurement that tests whether this makes it more effective.

A Inspection routes and reproducibility

Question	Start here
How does the repository fit together?	docs/ARCHITECTURE.md
What is proved and what is open?	docs/RESULTS.md and docs/claims.json
Which declaration states each paper result?	docs/paper_lean_coverage.json
Where is the checked mathematics?	lean/Erdos249257.lean and lean/ErdosProblems.lean
What does Comparator check?	docs/EXTERNAL_VERIFICATION.md , verification/comparator.json and the corpus replay index evidence/comparator/associations.json
Whose earlier work does each result use?	docs/PRIOR_ART.md
Which checks gate a release?	scripts/check_release.py and .github/workflows/lean.yml
How can work return with credit?	CONTRIBUTING.md , credit policy and architecture contributions
Which agent skills exist?	skills/README.md

Table 4: Entry points for inspection. They lead to the authority-bearing files and add no authority of their own.

A fresh clone can inspect the control card and check its committed navigation without Lean:

```
python3 scripts/proof_cockpit.py --format card
python3 scripts/proof_cockpit.py --check
```

Release-surface validation is a separate operation using the pinned Python requirements in [the reproduction guide](#); it runs `python3 scripts/check_release.py`. Formal replay additionally uses the pinned Lean and Comparator environment named by the card and guide. The paper inventory, `docs/publication_contract.json`, records source and PDF hashes and validation commands, and the evidence for the checker study in Section 4 is recorded in `docs/publication_evidence.json`.

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